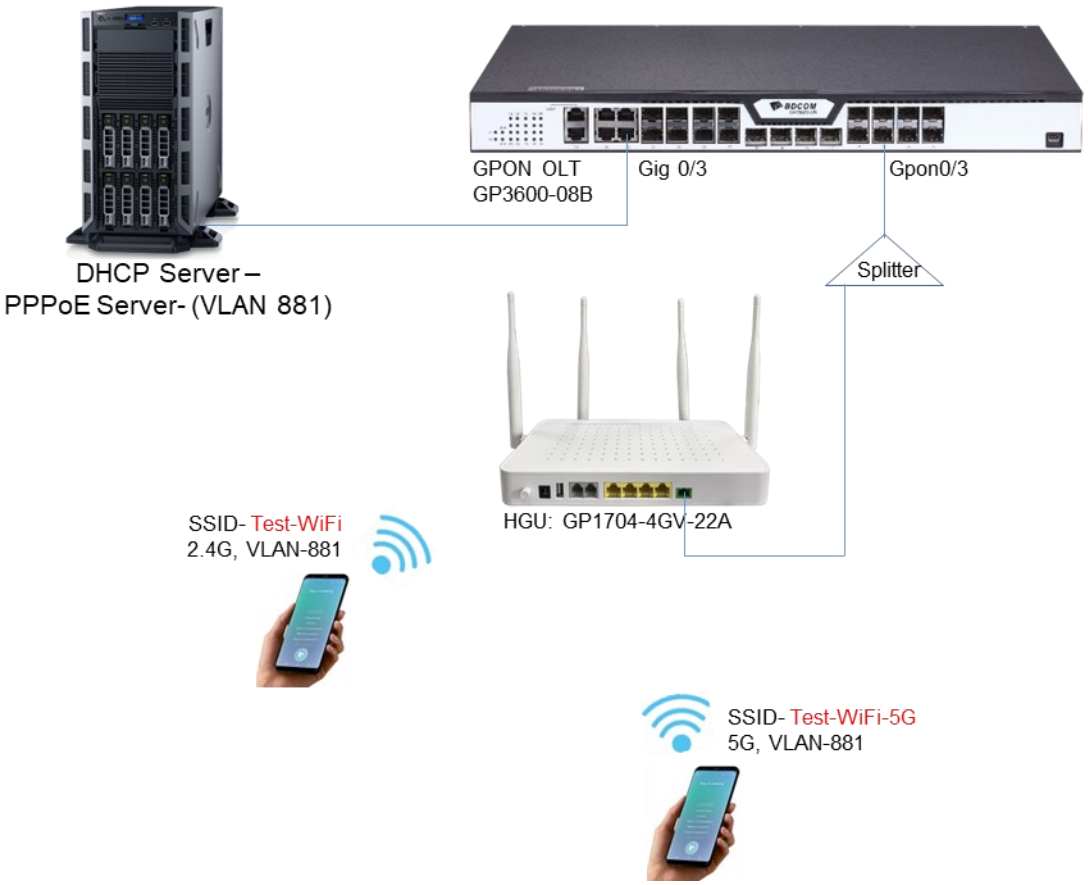
Configuring HGU ONU from OLT side through command and from OLT web interface	
OLT Model	GP3600-08B
Firmware Version	Build 106034
HGU ONU Model	GP1704-4GV-22A
ONU Firmware Version	10.0.33D.1324



The Topology Diagram

Uplink Configuration –

```
GP3600-08B#
GP3600-08B#show running-config interface gigaEthernet 0/3
Building configuration...

Current configuration:
!
interface GigaEthernet0/3
 description Uplink_S2528-B
 switchport mode dot1q-tunnel-uplink
 no storm-control broadcast threshold
 no storm-control multicast threshold
 no storm-control unicast threshold
GP3600-08B#
```

PON port Configuration –

```
GP3600-08B#
GP3600-08B#show running-config interface gpoN 0/3
Building configuration...

Current configuration:
!
interface GPON0/3
 gpon bind-onutype onutype-default-hgu precedence 127
 gpon bind-onutype onutype-default precedence 128
 gpon bind-onu sn BDCM:B1E2A780 10
 switchport trunk vlan-allowed 881
 switchport mode trunk
 switchport protected 1
 storm-control broadcast threshold 5
 storm-control multicast threshold 5
 storm-control unicast threshold 5
 gpon transceiver Any-Reset-Preamble
GP3600-08B#
```

Need to check the ONU status if it is "**Active Success**" mode. If the ONU is successfully registered with the OLT we may proceed to the next step to configure ONU through command line.

Here, we will configure gpon0/3:10 (which means the ONU is connected form PON3 and the ONU sequence is number 10). First lets see the status of ONU WAN interface by the commads –

Show gpon interface gpoN 0/3:10 onu wan 1 config
Show gpon interface gpoN 0/3:10 onu wifi 1 config

```
BDCOM(tm) GP3600-08B Software, Version 10.3.0D Build 106034
Copyright by Shanghai Baud Data Communication CO. LTD.
Compiled: 2022-11-17 14:16:4 by SYS, Image text-base: 0x108000
ROM: System Bootstrap, Version 0.1.7, Serial num:00324001729
System image file is "flash:/switch.bin"
hardware version:A
(RISC) processor with 1048576K bytes of memory, 32768K bytes of flash
Base ethernet MAC Address: 98:45:62:30:85:8c
PCB version:A
CPLD version:A
snmp info:
  product_ID:405   system_ID:1.3.6.1.4.1.3320.1.405.0
GP3600-08B uptime is 0:23:31:19, The current time: 2022-11-25 16:14:20
 Reboot history information:
  No. 1: System is rebooted by power-on
  No. 2: System is rebooted by command at 2000-1-6 19:54:40, uptime 5:19:54:04
GP3600-08B#
GP3600-08B#show gpon onu-information interface gpon 0/3 10
Interface GPON0/3 has bound 1 ONUs:
IntfName       VendorID  ModelID      SN               LOID                    Status   Config Status Active Time
-------------- --------- ------------ ---------------- ----------------------- -------- ------------- -------------------
GPON0/3:10     BDCM      GP1705-4GVR  BDCM:B1E2A780    N/A                     active   success       2022-11-25 16:12:39

GP3600-08B#show gpon onu-image-information interface gpon 0/3:10
Image information of GPON0/3:10:
 Image#0  10.0.33D.1324   valid    active    committed
 Image#1  10.0.33D.1055   valid    inactive  uncommitted

GP3600-08B#show running-config interface gpon 0/3:10
Building configuration...

Current configuration:
!
interface GPON0/3:10
 gpon onu model-id GP1705-4GVR
 gpon onu tcont-virtual-port-bind-profile tvbind-default
 gpon onu flow-mapping-profile flow-mapping-default-hgu
 gpon onu virtual-port 1 gem-port 264
 gpon onu tcont 1 alloc-id 261
 gpon onu catv 1 disable
GP3600-08B#
```

We won't be able to see any configuration or any IP address or any SSID because we didn't configure the ONU yet.

```
GP3600-08B#show gpon interface gpoN 0/3:10 onu wan 1 config
        diID 213 WAN 1:
        adminStatus                     enable
        linkStatus                      DOWN
        NATState                        disable
        connectionType                  DHCP
        serviceType                     tr069
        MTU                             1500
        autoGetIPAddressMode            enable
        TCIVlan                         46
        TCIPriority                     7
        TCIDei                          0
        dialMode                        always
        dialOnDemandTimeout             0
        address                         0.0.0.0
        mask                            0.0.0.0
        gateway                         0.0.0.0
        primaryDNS                      0.0.0.0
        secondaryDNS                    0.0.0.0
        bindPort                        none
        pppoe_username
        pppoe_password
        connection name                 1_TR069_R_VID_46
        PPPoE service name
        ipv6 address                   ::
        ipv6 mask                      ::
        ipv6 gateway                   ::
        ipv6 primaryDNS                ::
        ipv6 secondaryDNS              ::
        auto get ipv6 address          disable
        auto get DNS address           enable
        DHCPV6 prefix delegation       disable
        LAN DHCP                       enable
        ip version                     IPv4 Only
```

```
GP3600-08B#show gpon interface gpoN 0/3:10 onu wifi 1 config
         ssid                                    1
         mode                                    2.4G
         Channel id                              auto
         Channel width                           20MHz
         Tx power mode                           100%
         WIFI hardwareVersion                    8192FR
         WIFI softwareVersion                    3.6.4
         ssid item                               4
         valid ssid                              1 2 3 4
         Txrate                                  auto
         ctrlBand                                upper
         wlanEnable                              enable
         ssidName                                2.4G-A780
         ssidAdmin                               enable
         ssidBroadcastAdmin                      enable
         ssidMaxsta                              0
         wlanstandard                            802.11b+802.11g+802.11n
         encryptMode                             WPA_WPA2
         encryptKey                              B1E2A780
         SGI                                     enable
```

Let's configure the HGU ONU gpon0/3:10 by using following commands –

GP3600-08B#config
GP3600-08B_config# interface gpoN 0/3:10
GP3600-08B_config#gpon onu wan 1
GP3600-08B_config#gpon onu wan 1 admin-status enable
GP3600-08B_config#gpon onu wan 1 tci vlan 881
GP3600-08B_config#gpon onu wan 1 service-type internet tr069
GP3600-08B_config#gpon onu wan 1 connection-type pppoe
GP3600-08B_config#gpon onu wan 1 ip-version ipv4
GP3600-08B_config#gpon onu wan 1 pppoe username test password test
GP3600-08B_config#gpon onu wan 1 nat enable
GP3600-08B_config#gpon onu wan 1 lan-dhcp enable
GP3600-08B_config#gpon onu wan 1 bind lan1 lan2 lan3 lan4 ssid1 ssid5

```
GP3600-08B_config#
GP3600-08B_config#interface gpoN 0/3:10
GP3600-08B_config_gpon0/3:10#
GP3600-08B_config_gpon0/3:10#gpon onu wan 1
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 admin-status enable
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 tci vlan 881
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 service-type internet tr069
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 connection-type pppoe
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 ip-version ipv4
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 pppoe username test password test
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 nat enable
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 lan-dhcp enable
GP3600-08B_config_gpon0/3:10#gpon onu wan 1 bind lan1 lan2 lan3 lan4 ssid1 ssid
GP3600-08B_config_gpon0/3:10#
GP3600-08B_config_gpon0/3:10#exit
GP3600-08B_config#
```

NB: Here the ONU WAN interface is configured under VLAN881 for PPPoE connectivity. The username is "test" and the password is "test". Service-type should be "internet tr069" because of administering the ONU from remote. "lan-dhcp" should be enabled so that the client (laptop/mobile etc.) can get the IP from ONU.

GP3600-08B_config#gpon onu wifi 2.4G ssid 1 ssid-name ***Test-WiFi***
GP3600-08B_config#gpon onu wifi 2.4G ssid 1 encrypt-mode wpa_wpa2
GP3600-08B_config#gpon onu wifi 2.4G ssid 1 encrypt-key 12345678

GP3600-08B_config#gpon onu wifi 5g ssid 5 ssid-name ***Test-WiFi-5G***
GP3600-08B_config#gpon onu wifi 5g ssid 5 encrypt-mode wpa_wpa2
GP3600-08B_config#gpon onu wifi 5g ssid 5 encrypt-key 12345678
GP3600-08B_config#exit

```
GP3600-08B_config_gpon0/3:10#gpon onu wifi ?
  2.4G  -- WIFI 2.4G mode config
  5G    -- WIFI 5G mode config
GP3600-08B_config_gpon0/3:10#
GP3600-08B_config_gpon0/3:10#gpon onu wifi 2.4G ssid 1 ssid-name Test-WiFi
GP3600-08B_config_gpon0/3:10#gpon onu wifi 2.4G ssid 1 encrypt-mode wpa_wpa2
GP3600-08B_config_gpon0/3:10#gpon onu wifi 2.4G ssid 1 encrypt-key 12345678
GP3600-08B_config_gpon0/3:10#
GP3600-08B_config_gpon0/3:10#gpon onu wifi 5g ssid 5 ssid-name Test-WiFi-5G
GP3600-08B_config_gpon0/3:10#gpon onu wifi 5g ssid 5 encrypt-mode wpa_wpa2
GP3600-08B_config_gpon0/3:10#gpon onu wifi 5g ssid 5 encrypt-key 12345678
GP3600-08B_config_gpon0/3:10#
GP3600-08B_config_gpon0/3:10#exit
GP3600-08B_config#
```

Now let's check the ONU WAN interface and WiFi IP address and status by using the command –
Show gpon interface gpoN 0/3:10 onu wan 1 config

```
GP3600-08B#show gpon int gpoN 0/3:10 onu wan 1 config
        diID 213 WAN 1:
        adminStatus                     enable
        linkStatus                      UP
        NATState                        enable
        connectionType                  PPPOE
        serviceType                     internet tr069
        MTU                             1492
        autoGetIPAddressMode            enable
        TCIVlan                         881
        TCIPriority                     7
        TCIDei                          0
        dialMode                        always
        dialOnDemandTimeout             0
        address                         192.168.76.150
        mask                            255.255.255.255
        gateway                         192.168.76.1
        primaryDNS                      103.60.172.30
        secondaryDNS                    8.8.8.8
        bindPort                        LAN1 LAN2 LAN3 LAN4 SSID1 SSID5
        pppoe_username                  test
        pppoe_password                  test
        connection name                 1_TR069_Internet_R_VID_8
        PPPoE service name
        ipv6 address                   ::
        ipv6 mask                      ::
        ipv6 gateway                   ::
        ipv6 primaryDNS                ::
        ipv6 secondaryDNS              ::
        auto get ipv6 address          disable
        auto get DNS address           enable
        DHCPV6 prefix delegation       disable
        LAN DHCP                       enable
        ip version                     IPv4 Only
```

NB: We can see the ONU gpon 0/3:10 WAN interface gets PPPoE IP **192.168.76.150** along with subnet mask, Gateay and DNS from the server

We can see the ONU gpon 0/3:10 WiFi 2.4G status by using the command -
Show gpon interface gpoN 0/3:10 onu wifi 1 config

```
GP3600-08B#show gpon interface gpoN 0/3:10 onu wifi 1 config
        ssid                                    1
        mode                                    2.4G
        Channel id                              auto
        Channel width                           20MHz
        Tx power mode                           100%
        WIFI hardwareVersion                    8192FR
        WIFI softwareVersion                    3.6.4
        ssid item                               4
        valid ssid                              1 2 3 4
        Txrate                                  auto
        ctrlBand                                upper
        wlanEnable                              enable
        ssidName                                Test-WiFi
        ssidAdmin                               enable
        ssidBroadcastAdmin                      enable
        ssidMaxsta                              0
        wlanstandard                            802.11b+802.11g+802.11n
        encryptMode                             WPA_WPA2
        encryptKey                              12345678
        SGI                                     enable
```

We can see the ONU gpon 0/3:10 WiFi 5G status by using the command -

Show gpon interface gpoN 0/3:10 onu wifi 5 config

```
GP3600-08B#show gpon interface gpoN 0/3:10 onu wifi 5 config
        ssid                                    5
        mode                                    5G
        Channel id                              auto
        Channel width                           80MHz
        Tx power mode                           100%
        WIFI hardwareVersion                    8812FR
        WIFI softwareVersion                    3.6.4
        ssid item                               4
        valid ssid                              5 6 7 8
        Txrate                                  auto
        ctrlBand                                Auto
        wlanEnable                              enable
        ssidName                                Test-WiFi-5G
        ssidAdmin                               enable
        ssidBroadcastAdmin                      enable
        ssidMaxsta                              0
        wlanstandard                            802.11a+802.11n+802.11ac(5G)
        encryptMode                             WPA_WPA2
        encryptKey                              12345678
        SGI                                     enable
```

We can check the MAC address table in the OLT for VLAN 881 by the commad –

Show mac address-table dynamic vlan 881

Here we can see MAC from both uplink side (g0/3) and from ONU (gpon0/3:10) side.

```
GP3600-08B#show mac address-table dynamic vlan 881
        Mac Address Table (Total 2)
-----------------------------------------

Vlan    Mac Address       Type       Ports
----    -----------       ----       -----
881     0055.b1e2.a78a    DYNAMIC    gpon0/3:10-1
881     cc2d.e040.a22c    DYNAMIC    g0/3
GP3600-08B#
```

We can see VLAN in both uplink and PON port by the commands –

Show vlan interface gigaEthernet 0/3
Show vlan interface gpoN 0/3

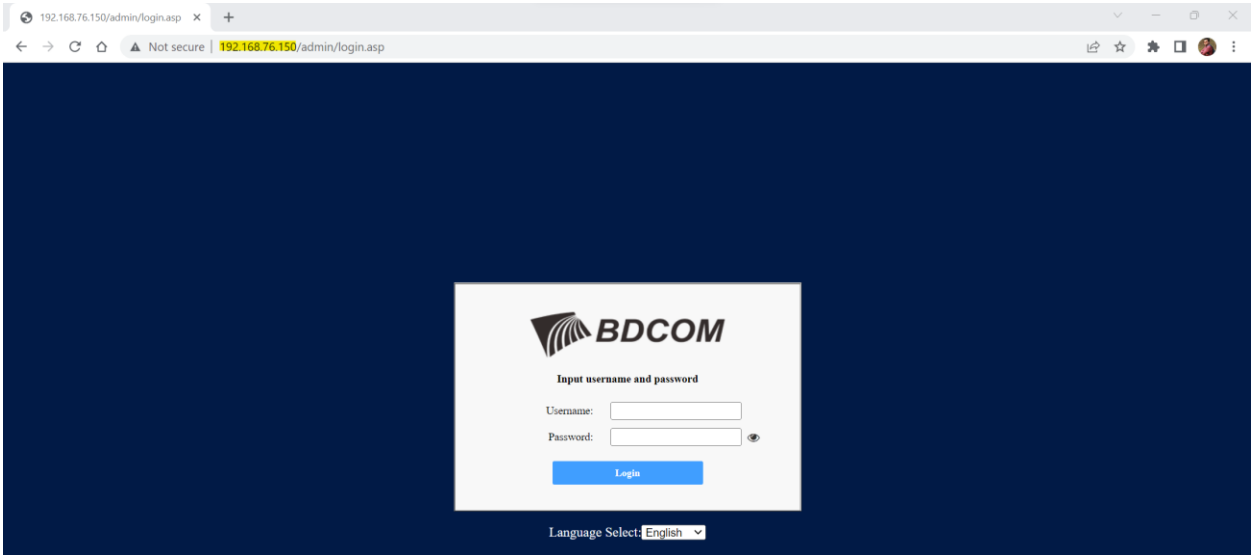
```
GP3600-08B#show vlan interface gigaEthernet 0/3

Interface            VLAN
Name                 Property PVID Vlan-Map         uTagg-Vlan-Map
-------------------- -------- ---- ---------------- ----------------
GigaEthernet0/3      Trunk    1    1,10-11,100,200  1
                                   300,500,881,1301
GP3600-08B#
GP3600-08B#show vlan interface gpON 0/3

Interface            VLAN
Name                 Property PVID Vlan-Map         uTagg-Vlan-Map
-------------------- -------- ---- ---------------- ----------------
GPON0/3              Trunk    1    881              none
GP3600-08B#
```

The WAN and WiFi configuration is done ! By default the ONU is not reachable to it's WAN interface. So, it needs to enable ping, telnet and web access to the ONU so that we may access to the ONU. Need to use following commands -

GP3600-08B#config
GP3600-08B_config#interface gpoN 0/3:10
GP3600-08B_config#gpon onu remote-access ping enable
GP3600-08B_config#gpon onu remote-access telnet enable
GP3600-08B_config#gpon onu remote-access webUI enable
GP3600-08B_config#exit

```
GP3600-08B#ping 192.168.76.150
PING 192.168.76.150 (192.168.76.150): 56 data bytes
.....
--- 192.168.76.150 ping statistics ---
5 packets transmitted, 0 packets received, 100% packet loss
GP3600-08B#
GP3600-08B#telnet 192.168.76.150
Connecting to remote host...
Press 'q' or 'Q' to quit connection.
Closing telnet connection...
GP3600-08B#
GP3600-08B#config
GP3600-08B_config#int gpoN 0/3:10
GP3600-08B_config_gpon0/3:10#gpon onu remote-access ping enable
GP3600-08B_config_gpon0/3:10#gpon onu remote-access telnet enable
GP3600-08B_config_gpon0/3:10#gpon onu remote-access webUI enable
GP3600-08B_config_gpon0/3:10#exit
GP3600-08B_config#exit
GP3600-08B#
GP3600-08B#ping 192.168.76.150
PING 192.168.76.150 (192.168.76.150): 56 data bytes
!!!!!
--- 192.168.76.150 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0/0/0 ms
GP3600-08B#
GP3600-08B#telnet 192.168.76.150
Connecting to remote host...
Press 'q' or 'Q' to quit connection.

Please enter the username and password.


Username:admin

Password:*********
# exit


 Connection closed by remote host.
GP3600-08B#
```

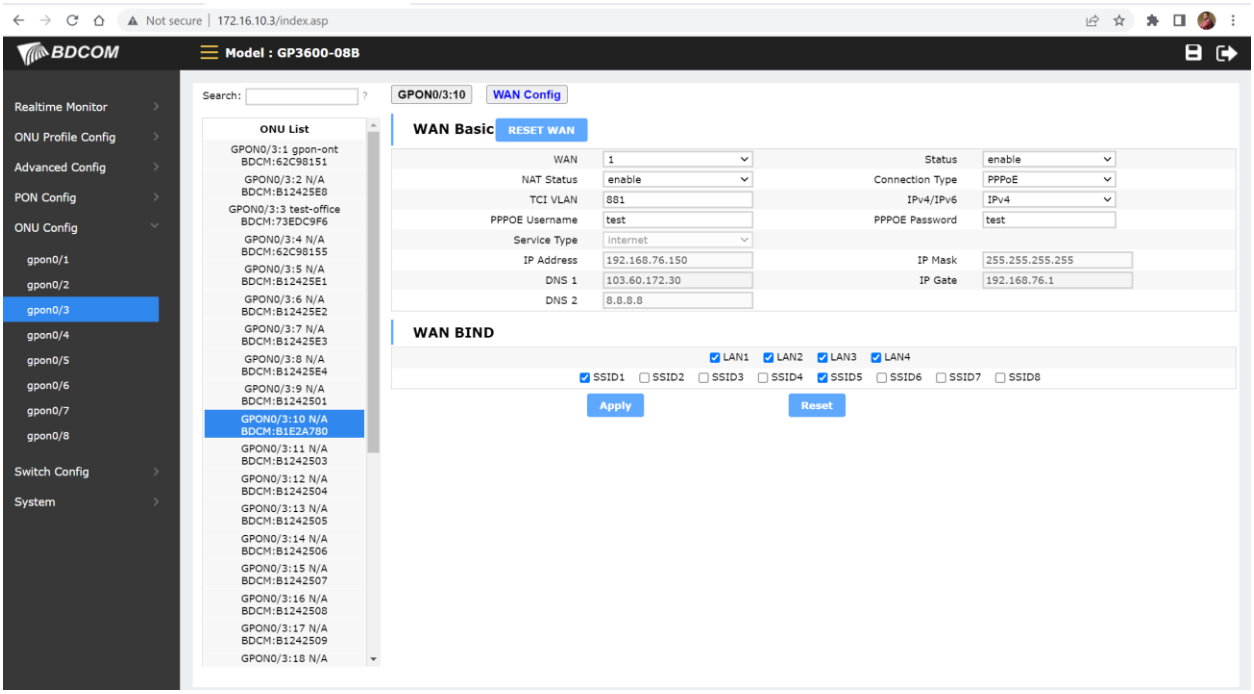
To access to the HGU ONU (gpoN 0/3:10) through web browser, we may browse the IP **192.168.76.150** which PPPoE IP ONU gets from PPPoE server



Configuring HGU ONU from OLT web interface

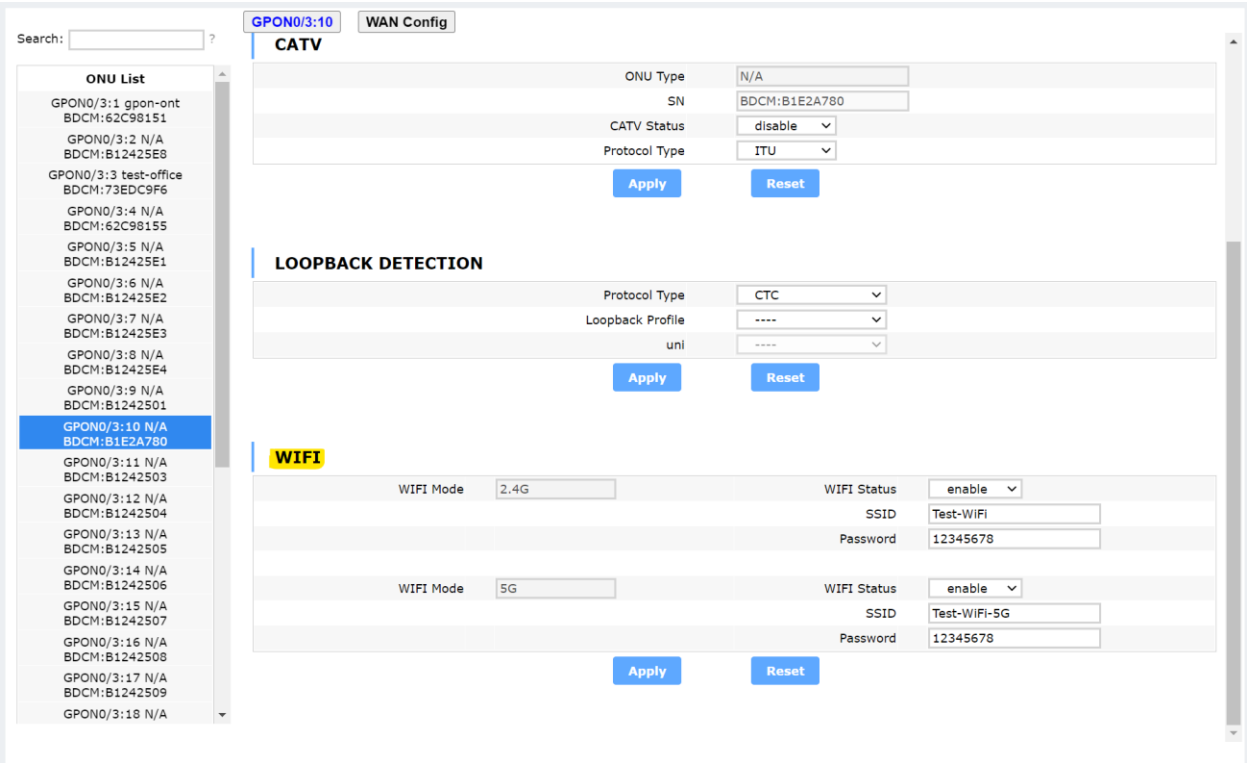
Access to the OLT by it's IP. Here the OLT IP is 172.16.10.3

Navigate to **ONU Config -> gpon0/3** and select the ONU **GPON0/3:10** and click on **WAN Config**



From here we can select Connection Type (PPPoE or DHCP or Static), username and password etc.

Navigate to **ONU Config -> gpon0/3** and select the ONU **GPON0/3:10** and click on **GPON0/3**



From here we can configure SSID, password for both 2.4G and 5G

-0-

Thank You